

Technical Lab Report: Digital Investigation & Network Forensics

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1. Objective

The goal of this lab is to run a forensic analysis on a network traffic capture file named (Digital_Investigation_Task.pcapng). By looking closely at the packet data, this investigation aims to map out active network protocols, track down unencrypted web traffic, find details about the user's computer setup, and extract files directly out of the raw network stream.

2. Investigation Phase & Findings

Phase 1: Protocol Hierarchy Analysis

I started the analysis by checking the protocol hierarchy in Wireshark to get a high-level view of the network traffic. This step helps map out what types of protocols are running and how much data they are using across the network layers. Reviewing this statistical breakdown early on makes it easy to spot any strange or unexpected protocols before diving into deep packet inspection.

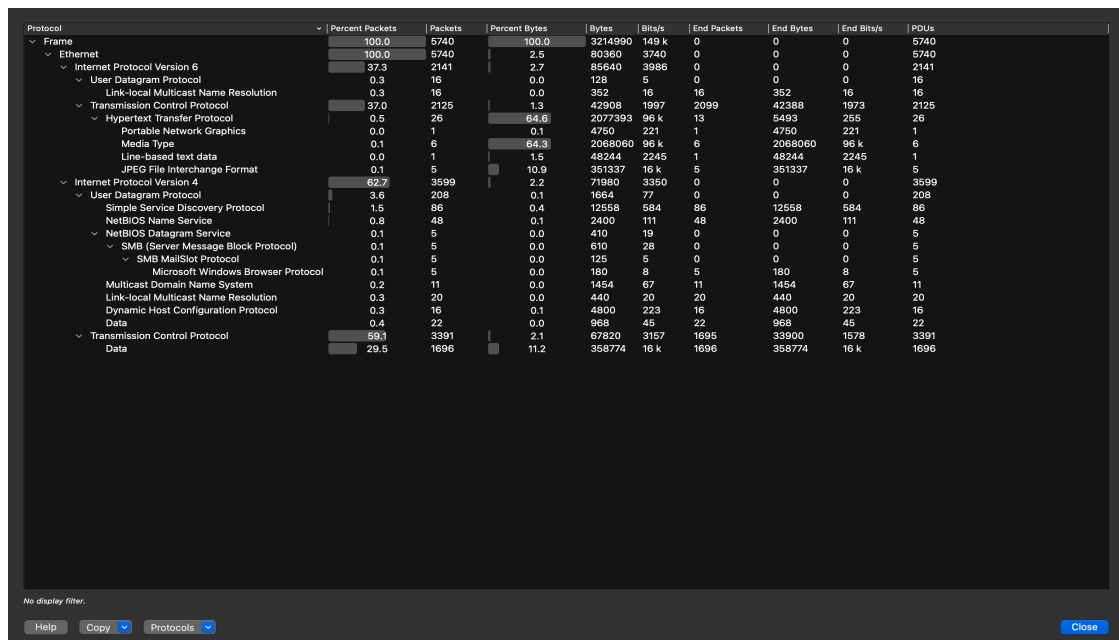


Figure 1: Statistical distribution of network protocols within the capture file.

Phase 2: Isolation of Unencrypted Web Traffic

Next, I used a display filter to look specifically for old HTTP traffic to see how secure data is while moving across the network. The filter brought up several plaintext packets, which proves that the user's computer was talking to external websites without any encryption. Looking at the info column, I could see standard web requests completely exposed, which makes the whole browsing session an easy target for packet sniffing or password stealing.

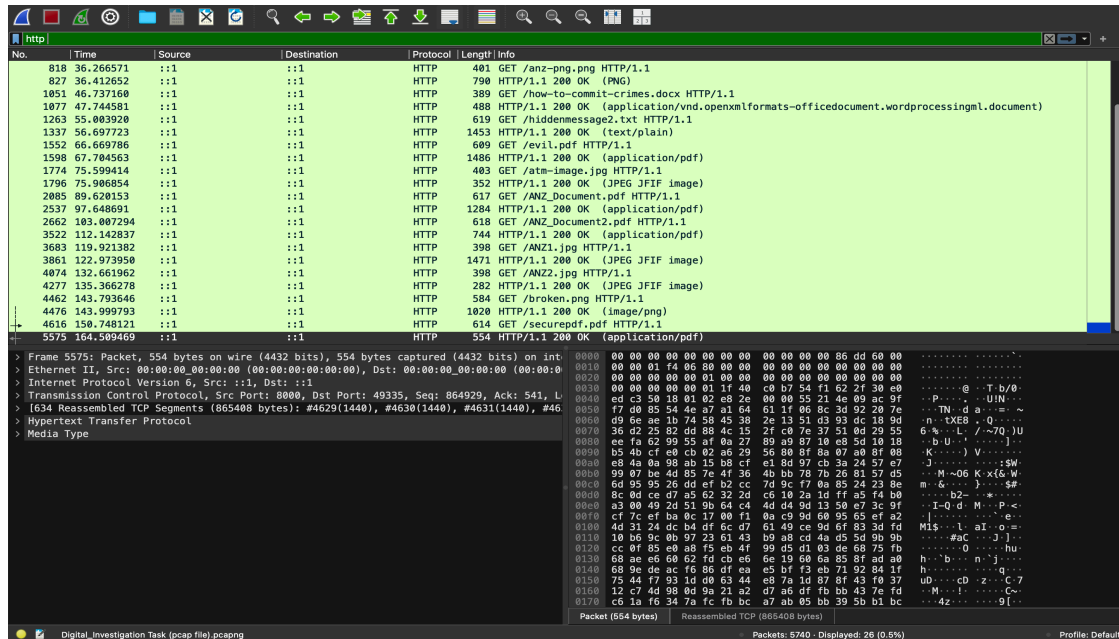


Figure 2: Filtered cleartext HTTP traffic indicating unencrypted transport.

Phase 3: Host Profiling via User-Agent Inspection

I dug deeper into the HTTP request headers to figure out what kind of device was being used on the network. By clicking on one of the inbound web packets and expanding the HTTP details section in Wireshark, I found the exact *User-Agent* string. This bit of metadata acts like a digital footprint, showing exactly what operating system and web browser version the target computer was running during the session.

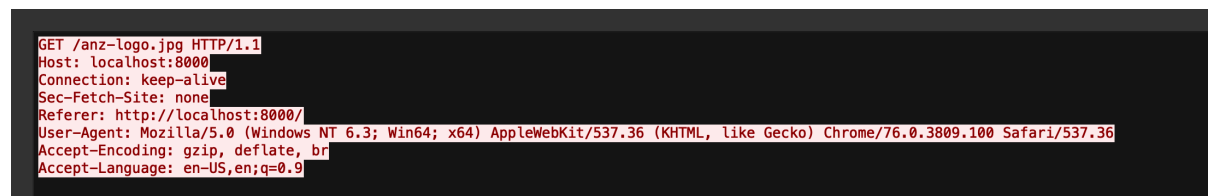


Figure 3: Extraction of host machine operating system and browser metadata.
[indicating windows NT 6.3 OS and Mozilla/5.0 web browser]

Phase 4: Forensic Object Extraction (File Carving)

Since the traffic was completely unencrypted, I was able to rebuild the files that were sent over the network. Instead of manually sorting through lines of raw hex data, I used Wireshark's object export feature to carve out the media file. I isolated the file pieces straight from the stream and saved the finished object directly to my computer, completely intact and uncorrupted.

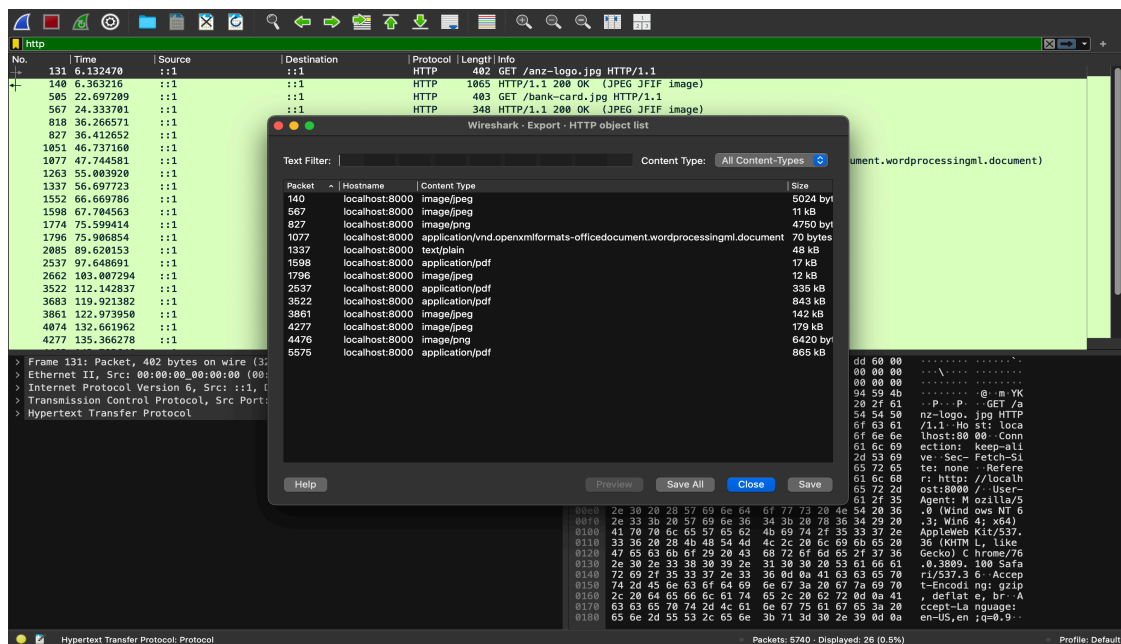
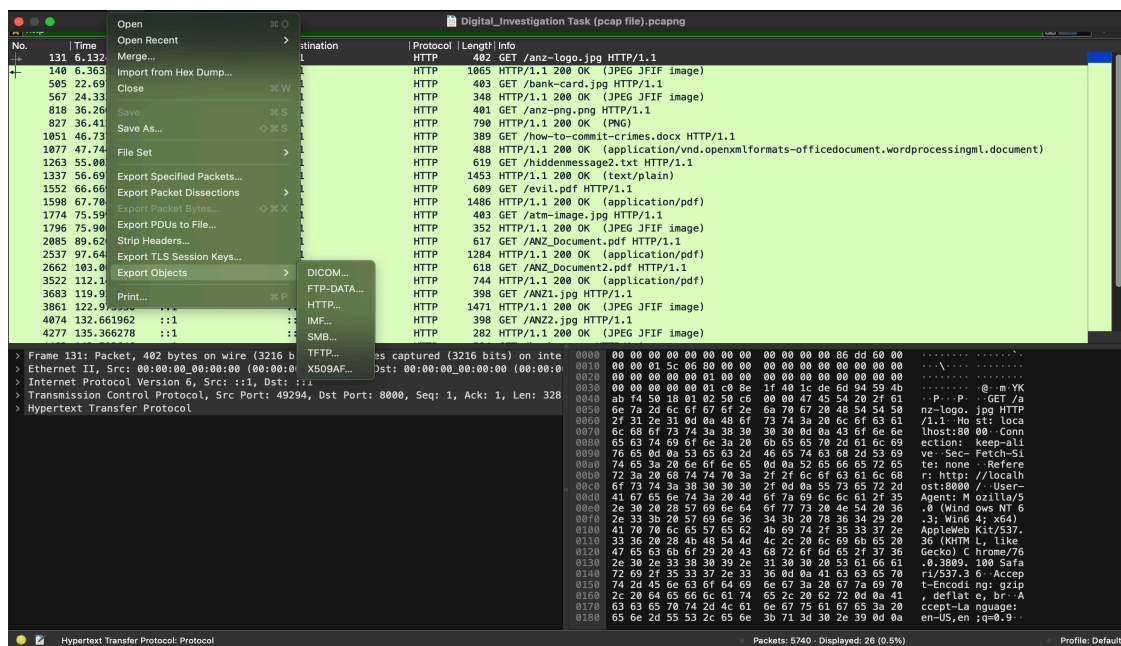


Figure 4: Forensic extraction of cleartext media objects from the network stream.

3. Conclusion & Engineering Remediation

The findings from this lab show that using old, cleartext protocols like HTTP leaves data completely exposed across the network. To protect this environment from passive sniffing and unauthorized file extraction, a couple of key security fixes should be applied:

- **Enforce HTTPS Everywhere:** Switch all web servers over to HTTPS using modern TLS 1.3 encryption. This makes sure that any data moving across the wire cannot be read in plaintext.
- **Set Up Port Security:** Use basic network access control features on your switches, like port security and VLAN separation. This stops unauthorized devices from sniffing traffic on the local network.